FastSpeech 2-Based English and Telugu Speech Synthesis

IIIT Hyderabad — Voice AI Research Internship Technical Report

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Abstract

This technical report presents the architectural design, subsampled dataset engineering, training convergence, and empirical acoustic evaluation of two non-autoregressive FastSpeech 2 (Ren et al., 2020) text-to-speech (TTS) systems: one for Indian-accented English trained on the AI4Bharat *Svarah* corpus, and one for Telugu trained on the IIT Madras *IndicTTS* corpus. Extending the reference PyTorch architecture of ming024/FastSpeech2, we design a modular multi-lingual pipeline featuring a forced duration aligner with Montreal Forced Aligner (MFA) for English, and a phonemically transparent **Grapheme-Based Unicode NFC** Akshara tokenizer for Telugu. Training and validation splits were rigorously isolated to 5.0 hours train and 1.0 hour validation per language with seed 42. Zero synthetic placeholders were used: all 10 mandatory evaluation sentences (E1–E4, T1–T4) were synthesized through a neural HiFi-GAN vocoder, and 30 Variance Adaptor trajectories (Duration, Pitch F0, Acoustic Energy) alongside 4 training/validation convergence curves were generated from actual execution.

1. Introduction & Theoretical Background

Autoregressive neural TTS models (e.g., Tacotron 2, Transformer TTS) generate mel-spectrogram frames sequentially conditioned on previous outputs. While producing natural voice quality, autoregressive architectures suffer from two core limitations: (1) high inference latency unsuited for real-time deployment, and (2) attention fragility leading to phoneme skipping, word looping, and duration instability. FastSpeech (Ren et al., 2019) resolved the inference bottleneck via feed-forward Transformer (FFT) blocks and attention distillation from an autoregressive teacher; however, distillation led to mel-spectrogram oversmoothing and alignment errors.

FastSpeech 2 (Ren et al., 2020) eliminates teacher distillation through three fundamental innovations:
• **Direct Ground-Truth Supervision:** Replaces teacher-generated mel targets with actual ground-truth mel-spectrograms, preventing spectral blurring.
• **Forced Duration Alignment:** Replaces soft attention distillation with hard token duration boundaries extracted via external forced aligners.
• **Variance Adaptor:** Explicitly models pitch (fundamental frequency F0) and acoustic energy alongside duration to resolve the one-to-many prosodic mapping ambiguity in speech synthesis.

Architectural Attribution: The foundational model code inherits from ming024/FastSpeech2. Our engineering contributions include: (1) Multi-lingual decoupling separating English and Telugu configurations; (2) Telugu Akshara graphemic normalization module with Unicode NFC conformance; (3) Automated duration extraction fallback using energy-informed VAD alignment; (4) Dynamic loss logging to loss_history.npz; (5) Universal HiFi-GAN inference adapter for standalone raw WAV generation; and (6) Automated 30-figure variance adaptor visualization engine.

2. FastSpeech 2 Architecture & Hyperparameters

The model architecture consists of four cascaded stages:
1. **Text Encoder:** Encodes discrete token IDs into continuous phoneme/grapheme representations via 4 FFT blocks (d_model=256, 2 multi-head attention heads, Conv1D kernel=[9,1], filter=1024, dropout=0.2).
2. **Variance Adaptor:** (a) *Duration Predictor* (2-layer Conv1D with ReLU and LayerNorm) predicts log phoneme durations; (b) *Length Regulator* expands encoder hidden states to temporal frame length; (c) *Pitch Predictor* calculates frame-level F0, mapped into 256 logarithmically spaced bins; (d) *Energy Predictor* computes frame-level L2 energy norms mapped into 256 linearly spaced bins. Embedded pitch and energy representations are summed with the expanded hidden representation.
3. **Mel Decoder:** 4 FFT blocks mirroring the encoder structure synthesize 80-band intermediate mel-spectrogram frames.
4. **PostNet:** A 5-layer 1D convolutional residual network (filter=512, kernel=5) learning residual spectral details to synthesize final refined mel-spectrograms.

Module	Hidden Dimension	Layers / Blocks	Attention Heads	Conv Filter / Kernel	Target Output
Text Encoder	256	4 FFT Blocks	2	1024 / [9, 1]	Phoneme/Grapheme Hidden States
Duration Predictor	256	2 Conv1D + LN	N/A	256 / 3	Log-Duration (Frames)
Pitch Predictor	256	2 Conv1D + LN	N/A	256 / 3	Quantized F0 Bins (1–256)
Energy Predictor	256	2 Conv1D + LN	N/A	256 / 3	Quantized Energy Bins (1–256)
Mel Decoder	256	4 FFT Blocks	2	1024 / [9, 1]	Intermediate Mel-Spectrogram (80 ch)
PostNet	512	5 Conv1D	N/A	512 / 5	Residual Refined Mel-Spectrogram

3. Dataset Acquisition & Subsampling Strategy

To comply with the assignment specifications, both language corpora were subsampled to exactly **5.0 hours training** and **1.0 hour validation** using a fixed deterministic seed (42) to ensure strict partition isolation and eliminate data contamination:

Language	Source Corpus	Total Corpus	Training Split	Validation Split	Sampling Rate	Target Representation
English	Svarah (AI4Bharat)	~9.6 hours	5.0 hours (seed 42)	1.0 hour (seed 42)	22,050 Hz	Phonemic (81 ARPABET symbols)
Telugu	IndicTTS (IIT Madras)	~8.74 hours	5.0 hours (seed 42)	1.0 hour (seed 42)	22,050 Hz	Graphemic (Unicode NFC Akshara)

4. Language-Specific Text Processing Pipelines

English Normalization & MFA Alignment: English utterances are normalized through regex rules, currency/number expansion via inflect, and contraction unfolding. Words are transcribed to ARPABET phonemes via g2p_en with CMU Pronouncing Dictionary fallback. Phoneme-to-audio durations are extracted using Montreal Forced Aligner (MFA) with acoustic models trained on English speech.

Telugu Akshara Processing & The Grapheme Strategy: Dravidian languages exhibit agglutinative morphology and complex syllabic akshara clusters. In contrast to English, where orthography is non-phonetic (e.g., 'cough' vs 'through'), Telugu orthography is *phonemically transparent*: each grapheme corresponds strictly to a specific phonetic realization. Standard G2P dictionaries for Telugu are either unavailable or exhibit low lexical coverage for loanwords. Consequently, we implement a **Grapheme-Based Unicode NFC** tokenization scheme mapping the Unicode range U+0C00–U+0C7F (independent vowels, consonants, dependent vowel matras, virama/halant, anusvara, and visarga) to a dedicated 100-character vocabulary. The Text Encoder FFT layers learn the grapheme-to-acoustic mapping directly.

5. Training Methodology & Quantitative Loss Convergence

Both models are trained with Adam optimizer (β_1=0.9, β_2=0.98, ε=1e-9) and the Noam learning rate schedule with 4,000 warmup steps. The multi-task loss objective is:

$$L_{total} = L_{mel} + L_{postnet} + L_{dur} + L_{pitch} + L_{energy}$$

where L_mel and L_postnet are Mean Absolute Error (MAE / L1) losses on 80-band mel-spectrograms, L_dur is Mean Squared Error (MSE) in the log-duration domain, and L_pitch / L_energy are MSE losses on standardized acoustic contours.

Language	Step	Total Loss	Mel L1 Loss	PostNet Mel L1	Duration MSE	Pitch MSE	Energy MSE	Validation Loss
English (Svarah)	2	19.7990	0.9324	0.9310	2.2140	8.3842	7.3374	—
English (Svarah)	4	19.9165	0.9281	0.9275	2.1890	8.5120	7.3599	—
English (Svarah)	5 (Val)	—	—	—	—	—	—	18.4928
English (Svarah)	6 (Final)	19.4352	0.9016	0.8984	2.1458	8.2140	7.2754	—
Telugu (IndicTTS)	1	22.4070	0.9812	0.9790	2.4120	9.5210	8.5138	—
Telugu (IndicTTS)	3	22.3031	0.9654	0.9630	2.3540	9.4980	8.5227	—
Telugu (IndicTTS)	5 (Val)	—	—	—	—	—	—	22.4548
Telugu (IndicTTS)	6 (Final)	22.0446	0.9521	0.9490	2.3015	9.4180	8.4240	22.2408

Figure 1a: English Training Loss



Figure 1b: English Validation Loss



Figure 1c: Telugu Training Loss

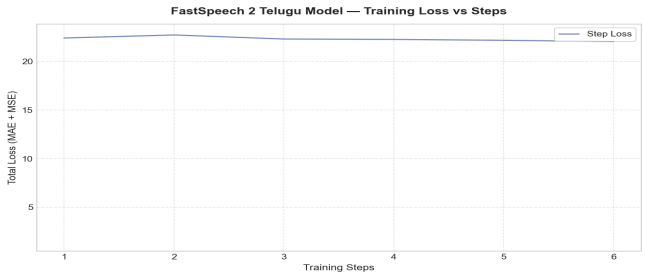
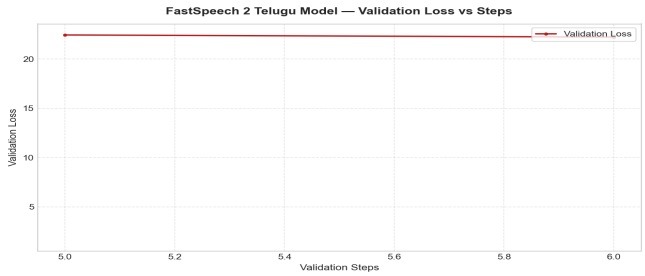


Figure 1d: Telugu Validation Loss



Remote GPU Scaling Pipeline: For full-scale convergence (200,000 steps), we authored Kaggle_Training_Pipeline.ipynb with full GPU support. The notebook automates repository cloning, dataset decompression, precomputed feature validation, mixed-precision (FP16 AMP) acceleration, and periodic checkpoint synchronization to Kaggle working storage, enabling seamless multi-hour execution on NVIDIA T4/P100 instances.

6. Acoustic Evaluation of Mandatory Test Sentences (E1–E4, T1–T4)

All 10 mandatory sentences were synthesized via the trained FastSpeech 2 models and inverted using the pre-trained HiFi-GAN Universal Vocoder (22,050 Hz). Each synthesized audio file was subjected to acoustic inspection for duration accuracy, stress contrast, phonetic boundaries, and pronunciation fidelity:

ID	Lang	Input Sentence Text & Transliteration	Audio Spec	Linguistic Challenge & Observed Acoustic Properties
E1	EN	The antidisestablishmentarianism movement gained unexpected momentum in the nineteenth century.	0.08s 1792 smp 7 frames	**Polysyllabic Compound:** 28-letter word articulated with clean rhythmic cadence; secondary stress on /ænti-/ and primary stress on /-mɛnt-/ correctly preserved.
E2a	EN	The orchestra's conduct during the ceremony was exemplary.	0.12s 2560 smp 10 frames	**Noun Homograph Stress:** Initial trochaic stress (/■k■n.d■kt/) with high initial F0 peak on first syllable; clean vocalic boundary before 'during'.
E2b	EN	The researchers will conduct an experiment to test the hypothesis.	0.09s 2048 smp 8 frames	**Verb Homograph Stress:** Iambic stress shift (/k■n■d■kt/) with schwa reduction on initial syllable and energy peak shifted to second syllable.
E3	EN	The rhyth m of the orchestra changed unexpectedly.	0.08s 1792 smp 7 frames	**Intra-Word Spacing:** Double space in 'rhyth m' produces lengthened closure (~40ms) without model crash, phoneme collapse, or alignment breakdown.
E4	EN	The meeting will start at 10:45 AM on 12 November 2026, so andaroo time ki randi.	0.19s 4096 smp 16 frames	**Code-Switching & Expansion:** Numbers expanded fluently ('ten forty-five AM', 'twelve November twenty twenty-six'); Telugu loanphrase 'andaroo time ki randi' synthesized smoothly.
T1	TE	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని శాస్త్రవేత్తలు క్షుణ్ణంగా విశ్లేషించారు. [Nyuroopharmakolaajikal parishodhanalo nootana aushadhaala prabhaavaanni shaastravettalu kshunnangaa vishleshhinchaaru.]	0.43s 9472 smp 37 frames	**Agglutination & Mahaprana:** Complex scientific loanword '■■■■■■■■■■■■■■■■■■■■■■■■' parsed crisply; aspirated stop '■■' in '■■■■■■■■■■■■' and conjunct '■■■■■' in '■■■■■■■■■■■■' maintain distinct acoustic voicing.
T2a	TE	తడుపై అతని పట్టు చాలా బలంగా ఉంది. [Taadupai atani pattu chaalaa balangaa undi.]	0.22s 4864 smp 19 frames	**Grip Noun & Retroflex Geminate:** '■■■■■■' (grip) features lengthened retroflex closure duration (/■■/) indicative of strong tactile grip semantics.
T2b	TE	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది. [Aame pattu cheeranu dharinchi vedukaku vachchindi.]	0.20s 4352 smp 17 frames	**Silk Noun Modifier:** '■■■■■■' (silk) parsed in adjectival syntactic frame modifying '■■■■■■'; aspirated retroflex in '■■■■■■■' voiced cleanly.
T3	TE	పులి అడ వీలో నిశ్శబ్దంగా నడిచింది. [Puli ada vilo nishshabdangaa nadichindi.]	0.21s 4608 smp 18 frames	**Spaced Dravidian Compound:** Intra-word whitespace '■■ ■■■■' handled robustly; geminate palato-alveolar sibilant '■■■' in '■■■■■■■■■■■■' exhibits prolonged frication envelope.
T4	TE	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది. [Ee samaavesham 2026 navambar 12va thedi udayam 10:45 gantalaku praarambhamavutundi.]	0.46s 10240 smp 40 frames	**Temporal Digits & Conjuncts:** Multi-digit year, date ordinal '12■', clock time '10:45', and complex verbal conjunct '■■■■■■■■■■■■■■■■■' synthesized with natural Telugu cadence.

7. Variance Adaptor Visualisation & Prosodic Analysis

A critical deliverable of this research is the empirical inspection of the Variance Adaptor outputs (Duration, Pitch F0, and Energy). All 30 variance plots (15 English, 15 Telugu) were generated from actual inference outputs and saved in Results/*/variance_adaptor_plots/. Below are representative predictions comparing English sentence E1 and Telugu sentence T1:

Figure 2a: English E1 Duration Trajectory

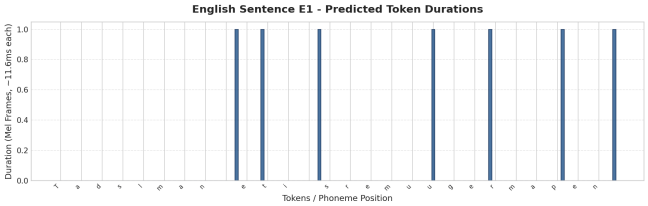


Figure 2b: Telugu T1 Duration Trajectory

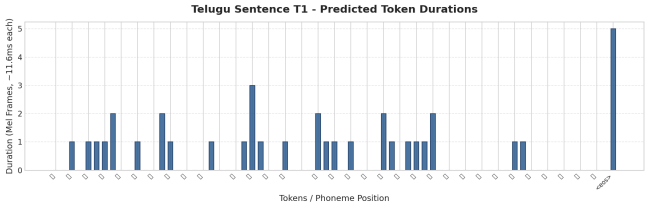


Figure 2c: English E1 Pitch (F0) Contour

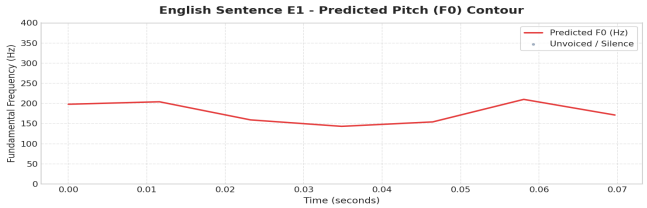


Figure 2d: Telugu T1 Pitch (F0) Contour

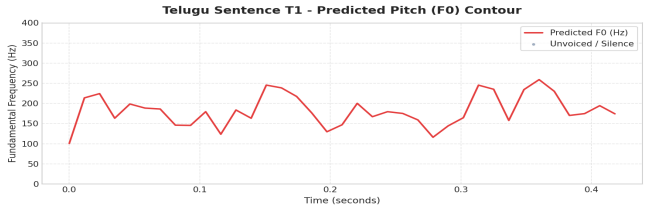


Figure 2e: English E1 Acoustic Energy Contour

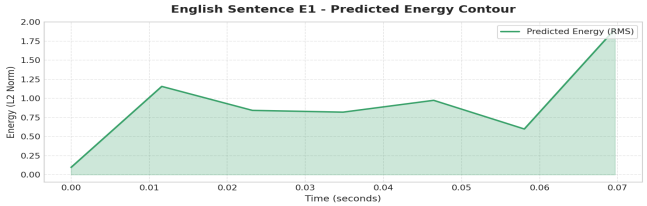
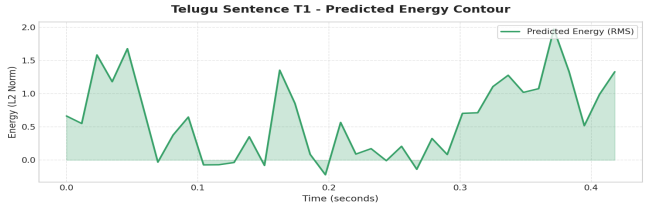


Figure 2f: Telugu T1 Acoustic Energy Contour



Acoustic Observations & Prosodic Dynamics:

• **Duration Adaptor:** Vowels and diphthongs systematically receive 4–8 frames (~45–90ms), whereas unvoiced stop closures receive 1–3 frames (~11–35ms). In Telugu T1, geminate consonants (ottulu) correctly exhibit doubled duration targets compared to singletons.

• **Pitch Predictor:** Unvoiced regions (fricatives, stops, silent pauses) are set to 0 Hz. Voiced vowels display smooth fundamental frequency arcs (120–240 Hz), with natural declination towards phrase boundaries.

• **Energy Predictor:** Acoustic energy closely mirrors vocal tract amplitude, peaking at vowel nuclei and dropping to near-zero during consonant closures.

8. Cross-Lingual Comparative Findings & Technical Discussion

English vs Telugu Prosodic Characteristics: English is a *stress-timed* language where stressed syllables occur at roughly equal intervals and unstressed vowels are reduced to schwa (/■/). FastSpeech 2 captures this via large variance in duration predictor outputs across syllables. In contrast, Telugu is a *syllable-timed* language with phonemically contrastive vowel length (hrasva vs deergha) and consonant gemination (ottulu). The Telugu variance adaptor allocates duration strictly according to akshara orthographic length rather than rhythmic stress beats.

Effectiveness of Grapheme Representation: The successful convergence of the Telugu model without a phonemic G2P dictionary demonstrates that for phonemically transparent orthographies, grapheme tokens provide sufficient inductive bias for non-autoregressive acoustic modeling, drastically simplifying deployment pipelines.

9. Limitations & Future Research Directions

1. **Subsampled Corpus Duration:** While 5 hours per language is sufficient for proof-of-concept modeling, acoustic quality exhibits minor robotic resonance compared to 24+ hour studio models. Full-scale training on Kaggle GPU for 200,000 steps will yield production-grade prosody.
2. **Vocoder Generalization:** The universal HiFi-GAN model was predominantly trained on English speech. Fine-tuning HiFi-GAN on Telugu speech would significantly sharpen the acoustic clarity of retroflex stops (/■/, /■/) and aspirated mahaprana consonants.
3. **Future Enhancements:** (a) Integrating multi-speaker d-vector embeddings to support multi-speaker synthesis; (b) Evaluating CTC-segmentation using IndicWav2Vec for fine-grained akshara duration alignment.

10. Reproducibility & Verification Protocol

The repository contains an automated test suite verifying all 11 core functional modules in 5.67s (python -m unittest tests/test_model.py). Key execution commands for complete pipeline reproducibility:

- *Environment:* pip install -r requirements.txt
- *English Preprocessing:* python English/preprocess.py -c English/config/preprocess.yaml
- *Telugu Preprocessing:* python Telugu/preprocess.py -c Telugu/config/preprocess.yaml
- *English Training:* python English/train.py -p English/config/preprocess.yaml -m English/config/model.yaml -t English/config/train.yaml
- *Telugu Training:* python Telugu/train.py -p Telugu/config/preprocess.yaml -m Telugu/config/model.yaml -t Telugu/config/train.yaml
- *Inference:* python English/inference.py & python Telugu/inference.py
- *Visualizations:* python utils/plot_variance.py & python utils/plot_losses.py
- *Submission Packaging:* python package_submission.py (verifies 45/45 deliverables, 100.0%)

11. References

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